

AI Design Document: Project Oracle

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Focus: Neural Ingestion & Predictive Modeling Layer

1. AI Strategy Overview

The Dante Labs AI layer for Project Oracle is built on a **Dual-Model Cognitive Engine**. It separates high-level logical reasoning (Language) from spatial market awareness (Vision). This document details the training, inference, and fine-tuning pipelines for both components.

2. NLP Module: PineCoder-70B

MODEL: TRANSFORMER-BASED LLM

A domain-specific Large Language Model optimized for **Pine Script v5** generation and logical trading strategy synthesis.

- **Base Model:** Llama-3-70B (Quantized 4-bit).
- **Dataset:** High-quality curated dataset of 120,000 TradingView scripts and the official Pine documentation.
- **Fine-Tuning:** LoRA (Low-Rank Adaptation) on NVIDIA H100 clusters.

2.1 Prompt Engineering & Control

```
<SYSTEM_INSTRUCTION>
Convert user intent into optimized Pine Script v5 code.
Rule: Use 'strategy()' for backtesting, 'indicator()' for visuals.
Edge-Case: Implement ATR-based risk management by default.
</SYSTEM_INSTRUCTION>
```

3. Computer Vision: ChartTopology-V1

MODEL: CONVOLUTIONAL NEURAL NETWORK (CNN) + VISION TRANSFORMER (ViT)

Responsible for "Topological Market Analysis"—interpreting the geometric structure of the chart directly from pixels.

Layer Type	Functionality	Output
Feature Extraction	Identifies candlestick clusters and volume density.	Tensor Representation
Pattern Head	Recognizes Wyckoff, Elliott Waves, and subjective S/R zones.	JSON Coordinates
Confidence Head	Predicts the probability of the pattern playing out.	Float (0.0 - 1.0)

4. Inference & Serving Pipeline

To support real-time TradingView overlays, the inference pipeline is optimized for **Latency over Throughput**.

- **Inference Engine:** NVIDIA Triton Inference Server.
- **Backend:** CUDA-accelerated Python environment with ONNX/TensorRT optimization.
- **Caching:** Redis caching for frequently requested pattern identifications on high-volume symbols (e.g., BTC, ETH).

5. Data Privacy & Model Security

Differential Privacy (DP)

User-specific strategy inputs are used for RAG (Retrieval-Augmented Generation) but are never used to re-train the global Dante Labs model to prevent cross-user "alpha leakage."

6. Model Evaluation (QA)

Metric	Target	Method
Script Syntactic Correctness	95%	Automated Pine Script Linter Validation.
Pattern Identification F1-Score	0.88	Backtesting against historical "perfect" pattern datasets.
Inference Speed	< 300ms	End-to-end gRPC measurement.

7. Future Roadmap: Reinforcement Learning (RLHF)

We plan to implement a feedback loop where user profitability (P&L) of AI-generated strategies serves as a reward signal to further optimize the PineCoder model's alpha generation capabilities.